INCH-POUND

MIL-H-43606C(GL)
10 December 1993
SUPERSEDING
MIL-H-43606B
10 June 1977

MILITARY SPECIFICATION

HAND CLEANER, WATERLESS

This specification is approved for use by the Natick Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of a waterless hand cleaner (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-C-96 - Cans, Metal, 28 Gage and Lighter

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5019 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8520

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STANDARDS

FEDERAL

FED-STD-595 - Colors Used in Government Procurement

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES (HHS), U.S. FOOD AND DRUG ADMINISTRATION (FDA)

Federal Hazardous Substances Act (16 CFR 1500)

Federal Food, Drug, and Cosmetic Act, and Regulations Promulgated Thereunder (21 CFR Parts 1-199)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 86 - Distillation of Petroleum Products
ASTM D 156 - Saybolt Color of Petroleum Products (Saybolt Chromometer Method)
ASTM D 287 - API Gravity of Crude Petroleum and Petroleum Products (Hydrometer Method)

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ASTM D 611 - Aniline Point and Mixed Aniline Point of Petroleum Products and Hydrocarbon Solvent
ASTM D 1266 - Sulfur in Petroleum Products (Lamp Method)
ASTM D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. All ingredients used in the manufacture of the hand cleaner shall be odorless, either white or colorless, of a type and grade applicable for use on the skin, and shall conform to the requirements of the Federal Food, Drug, and Cosmetic Act, and regulations promulgated thereunder (see 6.3). Kerosene chlorinated solvents, ammonia, free caustic, and abrasives are prohibited. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.2 Composition. The composition of the hand cleaner shall conform to the requirements of table I. Mixing directions shall be as specified in 6.4.

TABLE I. Composition

Ingredient	Percent by weight
Isoparaffinic hydrocarbon (technical)	31.0 ± 1.0
Light mineral oil (white, domestic, saybolt viscosity 125/135)	15.7 ± 0.5
Stearic acid (USP triple distilled)	8.0 ± 0.2
Mixed Isopropanolamine (technical)	4.0 ± 0.1
Distilled or deionized water	36.0 ± 1.0
Glycerine (USP)	3.0 ± 0.1
Sodium tetradecyl sulfate (technical)	1.0 ± 0.05
Polyvinylpyrrolidone (technical)	0.8 ± 0.05

3.2.1 Hydrocarbon solvent. The hydrocarbon solvent shall be an isoparaffinic hydrocarbon conforming to the following specific requirements when tested as specified in 4.2.4.

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	Minimum	Maximum
Specific gravity APR 60°F/60°F (15°C/15°C)	54.0	--
Color, Saybolt	30	--
Aniline point °F (°C)	180 (82)	--
Sulphur ppm	--	10
Distillation range °F (°C)	340 (171)	375 (191)

3.3 Color and texture. The hand cleaner shall be equal to or whiter than color chip No. 17886 of FED-STD-595 and shall be a smooth, homogeneous cream.

3.4 Stability (freeze-thaw and heat tests). The hand cleaner shall show no separation and shall develop no sharp color change or rancidity when tested as specified in 4.2.4.

3.5 pH. The hand cleaner shall have a pH of 5.5 to 8.0 when tested as specified in 4.2.4.

3.6 Odor. The hand cleaner shall leave no objectionable odor on the hands when tested as specified in 4.2.4.

3.7 Soil removal. The hand cleaner shall effectively remove soil and shall have a rating of "good" or "excellent" when tested as specified in 4.2.4.

3.8 Grit. The hand cleaner shall be free of grit when tested as specified in 4.2.4.

3.9 Labeling. Labeling shall conform to the regulations under the Federal Hazardous Substances Act. In addition to the regular marking, each unit container shall be durably and legibly marked with the following special marking:

Instruction for Use

1. Use sparingly for light soil. Place small amount of cleaner on the hands. Rub briskly until cleaner liquefies and dirt, grime, or smudges are loosened.

2. Wipe dry with paper towels or other available materials.

WARNING

Keep out of reach of children.
Do not take internally.
Do not use if hands or arms are susceptible to skin rash.

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3.10 Workmanship. The hand cleaner shall be clean and shall not contain dirt or foreign matter. In addition, the container shall not be dented, abraded, dirty, split, or punctured, and there shall be no seepage of contents.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.1.1 Certification. The contractor shall furnish a certificate of compliance certifying that the composition requirements specified in 3.2 have been met with the exception that the hydrocarbon solvent specified in 3.2.1 shall be tested as specified in 4.2.4.

4.2.2 End item visual examination. The end items shall be examined for the defects listed below. The lot size shall be expressed in units of hand

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cleaner unit containers. The sample unit shall be one unit container of hand cleaner. The inspection level shall be I and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Unit container:	
Construction	Not as specified (see 5.1)
Closure	Not as specified (see 5.1); imperfect
Workmanship	Dented, abraded, dirty, split, punctured; any seepage of contents
Markings	Missing, incomplete, incorrect, not in accordance with contract requirements Does not contain labeling information specified in 3.9
Contents:	
Color and texture	Color not as specified Contains lumps; not smooth, not homogeneous
Workmanship	Not clean, presence of dirt or foreign matter

4.2.3 End item net contents examination. The end items shall be examined for net contents. The lot size shall be expressed in units of filled hand cleaner unit containers. The sample unit shall be one filled unit container of hand cleaner. The inspection level shall be S-2. The lot shall be unacceptable if the average net contents per container, for all sample units examined, is less than required.

4.2.4 End item testing. The end items shall be tested for the characteristics listed in table II. The lot size shall be expressed in units of hand cleaner unit containers. Each lot shall consist of a single composition of product, manufactured under essentially the same conditions and at essentially the same time. The sample unit shall be one filled hand cleaner unit container. The sample size shall be as indicated in the table below. Any test failure shall be cause for rejection of the lot.

Lot size (number of unit containers)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

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4.2.5 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container
Content	Number per container is more or less than required

TABLE II. End item tests

	Specification reference		Requirements applicable to	Number determinations	Results reported as
Characteristic	Requirement	Test method	Individual unit	Per sample unit	Pass or fail 2/
Composition	3.2	1/			
Stability (freeze-thaw, and heat tests)	3.4	4.3.1 and 4.3.2	X	1	X
(pH)	3.5	4.3.3	X	2	X 3/
Odor	3.6	4.3.4	X	1	X
Soil removal	3.7	4.3.5	X	1	X
Hydrocarbon solvent	3.2.1	4.3.6	X	1	X
Grit	3.8	4.3.7	X	1	X

1/ The contractor shall submit to the contracting officer, or his authorized representative, production data certifying the compliance of each prepared batch with composition requirements (see 4.2.1.1).

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2/ If failure is indicated, report either description of failure or numerical point of failure, as applicable

3/ Both tests must pass this requirement.

4.3 Methods of inspection.

4.3.1 Freeze-thaw test. A 25 g ± 0.5 g sample of hand cleaner, in a 1 ounce (28 g) capped bottle, shall be exposed to a temperature of -40°F ± 2°F (-40°C ± 1°C) for 18 hours. The sample shall then be allowed to remain at room temperature until thawed. This test shall be repeated to five complete cycles. Any separation, rancidity, or sharp color change shall constitute failure of the test.

4.3.2 Heat test. A 25.0 g ± 0.5 g sample of hand cleaner, in a 1 ounce (28 g) capped bottle shall be exposed to a temperature of 155°F to 160°F (60°C to 71°C) for four hours. The sample shall then be allowed to cool to room temperature. This test shall be repeated for five complete cycles. Any separation, rancidity, or sharp color change shall constitute failure of this test.

4.3.3 pH. A small quantity of hand cleaner shall be applied to both sides of a strip of pH indicator test paper conforming to pH range 2 or 3 of acid full range pH kit (or equivalent), and the color in the paper allowed to develop. Compare the color of the test paper with the color of the color comparison chart. The indicated pH shall be in the range specified in 3.5. Any color outside the range shall constitute failure of the test.

4.3.4 Odor. A small quantity of the hand cleaner shall be examined to detect any objectionable odor. Aproximatley 1 g of the hand cleaner shall be rubbed on one hand and the excess removed with a paper towel. There shall be only a faint remaining pleasant odor on the skin. Any evidence of an objectionable odor shall constitute failure of the test.

4.3.5 Soil removal.

4.3.5.1 Soiling mixture. The test soil shall be prepared by mixing thoroughly the following ingredients (see 6.5):

1 quart (945 ml) SAE No. 20 nondetergent lubrication oil

1/2 pound (226 g) lime grease

4 ounces (113 g) carbon lamp black

6 ounces (170 g) charcoal black

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4.3.5.2 Soil removal test. Soil removal shall be determined as follows:

Step 1 — Wash both hands with soap and water, dry and wait 15 minutes.

Step 2 — Apply approximately 0.1 g of the soiling mixture to the back of the left hand and knuckles with the right forefinger. Rub the soiling mixture into the skin by stroking lengthwise, crosswise, and in a circular motion for one minute.

Step 3 — Remove any excess soiling mixture by wiping with a paper towel.

Step 4 — Rub 1 g of the hand cleaner under test into the soiled skin area for one minute. Wipe the excess cleaner and soil from the hand with a paper towel.

Step 5 — Repeat the operation described in step 4 and compare the treated hand with the unsoiled right hand. Rate the removal as excellent, good, fair, or poor as follows:

Excellent - soiling mixture is completely removed with no trace of skin discoloration.

Good - substantially all the soiling mixture is removed with soil remaining in the creases of the knuckles.

Fair - soil remains in both the pores and knuckle creases.

Poor - soil remains in pores, knuckle creases, and on skin surface.

A rating of fair or poor shall constitute failure of the test.

4.3.6 Hydrocarbon solvent. The specific properties of the isoparaffinic hydrocarbon shall be determined in accordance with the following ASTM test methods:

Property	Test methods
Specific gravity	ASTM D 287
Color, Syabolt	ASTM D 146
Aniline point ^{o}F (^{o}C)	ASTM D 611
Sulphur ppm	ASTM D 1266
Distillation range ^{o}F (^{o}C)	ASTM D 86

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Failure of one or more properties to meet the requirements in 3.2.1 shall constitute failure of the test.

4.3.7 Grit. Spread evenly approximately 1 gram of the hand cleaner on a clean microscope slide. Place a second clean microscope slide over the hand cleaner and press together with a rotary motion. After the cleaner is removed, any evidence of grit or any signs of scratches on either of the microscope slides shall constitute failure of the test.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A preservation. One pound (453 g) of hand cleaner shall be packaged in metal cans 404 x 308 round, conforming to type V, class 2 of PPP-C-96, except that terne plate, backplate, or 0.25 electrolytic tinplate shall not be used. The exterior coating of the cans shall conform to coating plan A of PPP-C-96. The interior coating of the cans shall conform to the protective coating requirement specified in PPP-C-96. The seaming compound shall not affect nor be affected by the cream hand cleaner or any of the liquid components of the hand cleaner. Alternatively to compound-lined double seamed ends and sides, the seams may be soldered.

5.1.2 Commercial preservation. The hand cleaner shall be preserved in accordance with ASTM D 3953.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. The hand cleaner, preserved as specified in 5.1, shall be packed in accordance with level A packing requirements for type V, friction plug cans in the appendix of PPP-C-96.

5.2.2 Level B packing. The hand cleaner, preserved as specified in 5.1, shall be packed in accordance with level B requirements for type V friction plug cans in the appendix of PPP-C-96.

5.2.3 Commercial packing. The hand cleaner, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Marking. In addition to any special marking required by the contract, interior packages and shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951 as applicable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. The hand cleaner covered by this specificaiton is intended for use by Civil Defense Agencies in survival kits. The hand cleaner is suitable for storage at extremely high and low ambient temperatures.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Levels of preservation and packing (see 5.1 and 5.2).

6.3 Federal regulation. Matters pertaining to compliance of the hand cleaner under Federal Food, Drug, and Costmetic Act should be referred to the Department of Health, Education, and Welfare, Food and Drug Administration, Rockville, MD 20852.

6.4 Mixing directions. The following mixing directions were followed:

Melt the stearic acid at 156°F to 158°F (69°C to 70°C). When melted, slowly add light mineral oil and isoparaffinic hydrocarbon. Then mix isopropanolimine, distilled water, glycerine, sodium tetradecyl sulfate, and polyvinylpyrrolidone and slowly add this mixture to the oil phase, with continuous stirring. After all ingredients have been combined, stir the emulsion until the ingredients are well blended. If stirred too vigorously or for a long period of time, air bubbles will develop and make it difficult or impossible to package 1 pound (453 g) of hand cleaner in a 1 pound (453 g) can.

6.5 Ingredients. Test soil ingredients were obtained from the following sources:

(a) SAE 20 nondetergent lubricating oil - Ashland Oil Company, Inc., P.O. Box 391, Ashland, KY 41101.

(b) Lime grease - Atlantic Lubricant 2, Atlantic Richfield Company, 260 South Broad Street, Philadelphia, PA 19101.

(c) Carbon lamp black - Lampblack 12, Cities Service Company, Columbian Division, 3200 West Market Street, Akron, OH 44313.

(d) Charcoal black - Forest Products Charcoal Company, 484 North Hollywood Street, Memphis, TN 38112.

6.6 Subject term (key word) listing.

Detergent
Soap

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6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:

Army - GL

Review activity:

Army - MD

Preparing activity:

Army - GL

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-H-43606C(GL)	1993 December 10

3. DOCUMENT TITLE
HAND CLEANER, WATERLESS

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON/DSN
U.S. Army Natick RD&E Center	508-651-4532	256-4532
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-IR Natick, MA 01760-5019	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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